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ORIGINAL ARTICLE

The role of clinical pharmacist in the management of resistant hypertension

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Abstract
Background Hypertension is a major contributor to global cardiovascular morbidity and mortality. Treatment-resistant hypertension (TRH) presents a significant management challenge, requiring a pharmacist-physician collaborative model to achieve sustained blood pressure (BP) control.
Aim This study aims to evaluate the impact of a clinical pharmacy interventions on BP control, medication adherence, and patient outcomes in patients with TRH in a primary care setting.
Methods A randomized controlled trial was conducted involving 142 patients with TRH at two primary care clinics. Participants were randomly assigned to either the intervention group, receiving clinical pharmacy services including medication reviews and personalized counseling, or the control group, receiving standard care. Medication adherence was assessed using the Medication Adherence Rating Scale, and patient outcomes, including BP control and quality of life (QoL), were measured over a 6-month period.
Results The intervention group demonstrated a significant improvement in medication adherence compared to the control group ($p < 0.001$), with an effect size of − 1.75. Clinical parameters, such as BP (mean reduction in BP: 27.9/13.4 mmHg), showed more favorable outcomes in the intervention group ($p = 0.003$). Furthermore, patients receiving clinical pharmacy services reported higher QoL scores ($p < 0.001$) and expressed higher satisfaction with medication management.
Conclusion Clinical pharmacy interventions significantly enhance medication adherence and improve patient achievement of target BP goals in TRH patients. Incorporating clinical pharmacy services into routine care can lead to better health management and increased patient QoL. Further research is needed to explore long-term benefits and cost-effectiveness.

Keywords Clinical pharmacy · Medication adherence · Patient outcomes · Randomized controlled trial · Treatment-resistant hypertension

Impact of findings on practice statements

- This study underscores the crucial role of clinical pharmacist-physician collaborative work in enhancing blood pressure (BP) control and overall outcomes for patients with treatment-resistant hypertension (TRH).
- The findings of significantly improved BP control, increased adherence to medications and lifestyle recommendations, higher patient satisfaction, and enhanced quality of life (QoL) underscore the importance of the pharmacist's role in TRH patients' care.
- By assessing the impact of pharmacist interventions, this research sheds light on the potential benefits of involving clinical pharmacists in the management of TRH.
- In a time when multidisciplinary healthcare teams are gaining prominence, this study contributes to the evolving understanding of how collaborative care models can improve patient's QoL and other outcomes in the context of TRH.

Extended author information available on the last page of the article

Introduction

Hypertension (HTN), a pervasive and often silent health concern, stands as a leading contributor to cardiovascular morbidity and mortality globally [1]. If left undetected and untreated, HTN significantly elevates the risk of severe complications, including myocardial infarction, heart failure, stroke, and, ultimately, mortality [2]. Antihypertensive medications, when appropriately employed, play a pivotal role in achieving blood pressure (BP) control and reducing the incidence of cardiovascular events [3].

Despite advancements in hypertension management, a subset of individuals presents a challenge known as treatment-resistant hypertension (TRH). This condition, as defined by the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood

Pressure (JNC-8) and American College of Cardiology/American Heart Association (2017 ACC/AHA) guidelines, occurs when patients fail to attain target BP levels despite adherence to a comprehensive antihypertensive regimen [4].

Over the last four decades, pharmacists have undergone a transformative shift from their conventional role of merely dispensing drugs to assuming a more comprehensive responsibility for patient care. This evolution is underscored by an increased focus on pharmaceutical care services, with the primary goal of augmenting patients' comprehension of their medical conditions and fostering better adherence to prescribed medications [5, 6]. This paradigm shift has seen clinical pharmacists seamlessly integrate into multidisciplinary healthcare teams, providing valuable expertise in evidence-based prescribing, offering targeted counseling, and cultivating collaborative partnerships with prescribing professionals [7, 8]. Through the delivery of pertinent drug-related information and active involvement in patients' education and counseling, clinical pharmacists play a pivotal role in significantly improving treatment outcomes, promoting greater adherence, and ultimately enhancing the overall quality of life for patients [9, 10].

To date, the efficacy of team-based care in the setting of TRH management is still developing [11, 12]. The effect of the physician-pharmacist collaborative model on the management of TRH was previously reported in a few studies of either cohort or case report design [13, 14]. However, no similar studies were conducted in the Middle East region, specifically in Jordan. Therefore, the aim of this randomized, controlled clinical trial was to investigate the impact of the clinical pharmacist and physician collaborative model on the management of TRH guided by home BP monitoring and to evaluate the effect of patients' education on achieving BP control in patients with TRH in Jordan.

Aim

The aim of this study is to evaluate the effectiveness of clinical pharmacy interventions in improving medication adherence and patient outcomes in TRH in a primary care setting.

Ethics approval

The study received approval from the School of Pharmacy and the School of Postgraduate Studies at the University of Jordan. Additionally, the study adhered to ethical guidelines and obtained approval from the Jordanian Royal Medical Services (RMS) Institutional Review Board (IRB), approval number: 1405/2020/1. All eligible participants in the study provided signed informed consent forms.

Methods

Study design

This prospective, randomized, controlled trial was conducted at the King Hussein Medical Center (KHMC), which is a military medical complex that has five hospitals, a prominent referral hospital in Jordan. It was conducted from September 2020 to January 2021. The study was designed to assess whether the inclusion of clinical pharmacists as a vital component of the healthcare team would lead to improved BP control in patients with TRH. The COVID-19 situation led to adjusted clinic follow-ups via face-to-face interviews and subsequent phone-based interactions.

Study participant selection

To evaluate the impact of clinical pharmacist interventions on optimizing BP control and to compare the proportion of patients achieving BP goals in both study groups, the sample size was estimated based on similar RCTs conducted in a similar setting [15, 16]. The sample size calculation was performed with the following parameters: 80% power, 95% confidence intervals, and a significance level of 5%, using the following formula:

$$n = \frac{2 \bullet \sigma^2 \bullet \left(Z_{\alpha/2} + Z_\beta\right)}{d^2}$$

A minimum sample size of 55 patients per group was determined.

Given the impact of the COVID-19 pandemic and concerns about further dropouts, an additional 30% over-recruitment was applied to account for these issues, resulting in a final target sample size of 148 patients. The study targeted adult patients with previously diagnosed TRH attending internal medicine outpatient clinics at KHMC Hospitals. Inclusion criteria encompassed individuals aged 18 years or older who were diagnosed with TRH, defined as SBP of 140 mm Hg or higher and DBP of 90 mm Hg or higher despite the administration of three or more antihypertensive agents, including a diuretic. Patients with pregnancy, lactation, ICU admission, hemodialysis, and inability to provide informed consent were excluded from the study. Clinical pharmacists provided pharmaceutical care, including patient evaluation, treatment-related problems (TRP) detection [17], and education. Additionally, 22 physicians were involved, and questionnaires assessed their attitudes and satisfaction with clinical pharmacist services [18].

Randomization

Randomization was executed using a computer-generated random sequence with 1:1 allocation. Patients who met the inclusion criteria were assigned to one of two groups: the intervention group or the control group.

Patients in the intervention group received comprehensive care that involved a clinical pharmacist as an integral part of the healthcare team. The clinical pharmacist's responsibilities included providing education on hypertension, antihypertensive medications, and lifestyle modifications. Additionally, the clinical pharmacist conducted medication management, assessing the appropriateness of the prescribed regimen, recommending changes when necessary, and ensuring adherence. The clinical pharmacist also addressed lifestyle modifications, offering guidance on dietary choices and exercise routines to optimize BP control.

Patients in the control group received traditional care provided by physicians only. This care primarily consisted of standard medical consultations and physician-prescribed treatments. These patients did not benefit from clinical pharmacist interventions but were managed solely by the treating physicians.

Data collection

Data collection was carried out in two essential phases: baseline and follow-up. At the beginning of the study, patients' demographics, medical histories, and clinical characteristics were recorded. Baseline BP measurements (including SBP and DBP readings) were obtained to evaluate initial BP levels. In addition, patient adherence to antihypertensive medications and lifestyle modifications was assessed through a Compliance Score Assessment Sheet [15, 19]. To gauge patient satisfaction, the Satisfaction with Information about Medicines Scale (SIMS) was employed. Each question on the SIMS was rated on a scale of 0 to 1, with higher scores indicating greater satisfaction, as adapted from Horne et al. (2001) [19]. The SIMS was translated into Arabic; the Arabic version was validated by the research team. Lastly, patients' QoL was measured using the Arabic version of the Short Form-36 (SF-36) questionnaire [20]. Follow-up assessments were conducted after 12 weeks, which included repeat BP measurements, evaluation of medication adherence and lifestyle modifications, and QoL assessments.

At the end of the study period, an electronic pretested questionnaire was sent to each of the 22 doctors who were responsible for patient care during the clinic visit to assess their attitudes and satisfaction regarding the services offered by the clinical pharmacist [18].

Data analysis

Data analysis was conducted using the IBM Statistical Package for Social Sciences (IBM SPSS), version 23.0. Statistical tests for inferential statistics were set at a 95% confidence level, and p-values < 0.05 were deemed statistically significant. Baseline demographics and clinical characteristics were expressed as percentages, frequencies, mean ± SD, and n (%). Comparisons between study groups utilized independent sample t-tests and chi-square tests for normally distributed characteristics, while paired sample t-tests and Mann–Whitney tests were employed for within-group and non-normally distributed variables. Independent sample t-tests were used to compare absolute decreases in SBP and DBP between study arms. Achievement of JNC-8 BP goals was evaluated with chi-square or Fisher exact tests based on expected frequencies.

Comparisons were made between the intervention group (clinical pharmacist-physician collaborative care) and the control group (physician-only care) using appropriate statistical tests such as chi-square tests, t-tests, and analysis of variance (ANOVA).

Results

Study participants

During the enrollment period, a total of 1600 patients attended the internal medicine outpatient clinics at KHMC of the RMS hospital, out of which 1222 patients had HTN. Among them, only 154 (12.6%) had TRH and subsequently met the inclusion criteria. Of those, 148 (96.0%) patients consented to participate and were included in the trial, with 72 (48.6%) in the intervention group and 76 (51.4%) in the control/usual care group (Fig. 1). By the end of the 12-week study period, four patients (0.05%) from the intervention group and two patients (0.03%) from the control group were lost to follow-up. The final analysis included 142 patients, with 68 (47.9%) in the intervention group and 74 (52.1%) in the control group, within the accepted range of the calculated sample size due to over-recruitment (Fig. 1).

Baseline patient demographics and clinical characteristics

The study included 142 participants, comprising 87 males (60.4%) and 55 females (38.2%), aged between 35 and 86 years, with an average age of 62.6 ± 10.8 years. The

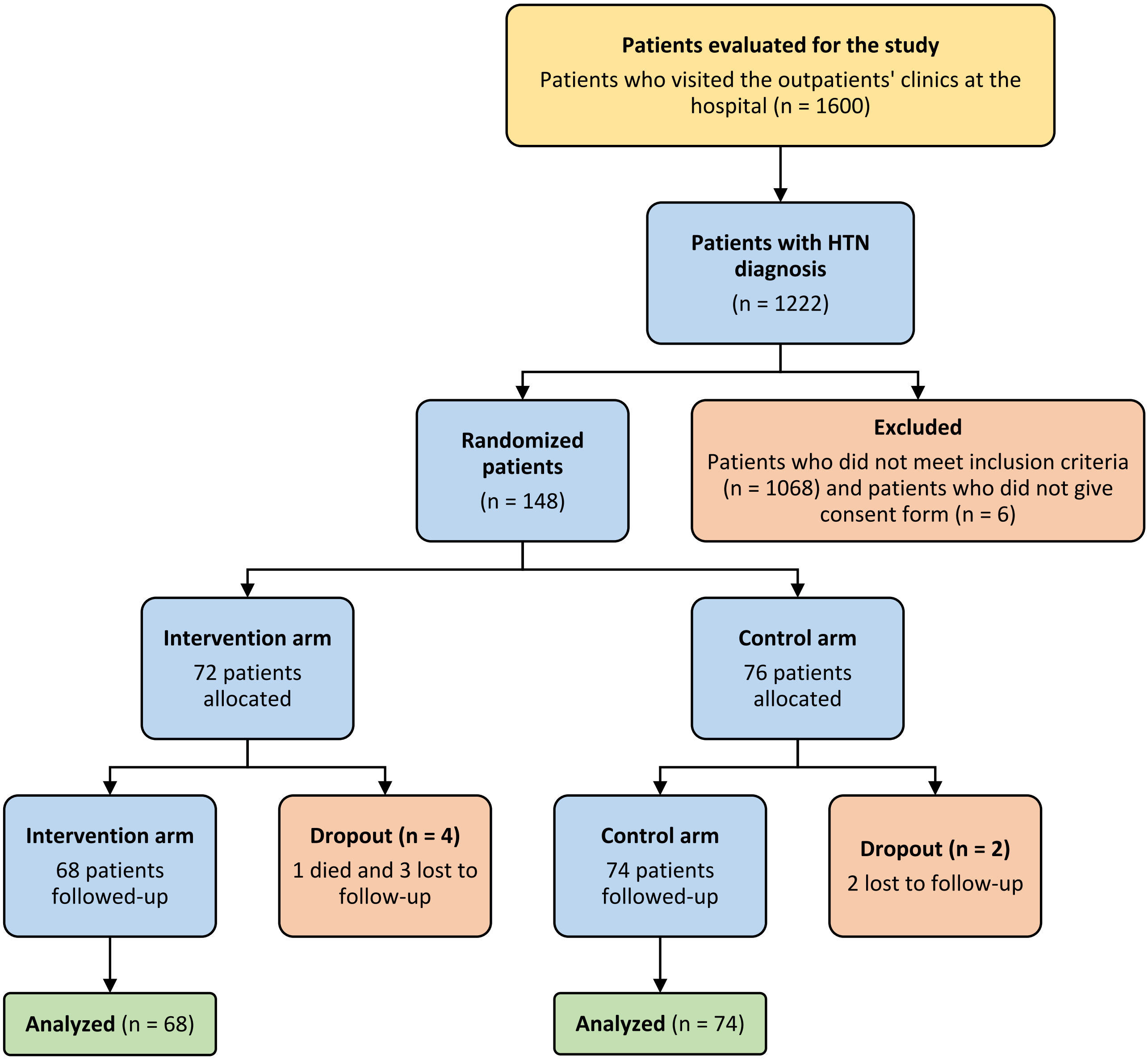

Fig. 1 Flow diagram of the study (CONSORT)

average number of medical conditions was 2.9 ± 1.16 for the intervention group and 2.8 ± 1.06 for the control group.

Apart from TRH, 74.6% of participants had ischemic heart disease, and 58.5% had diabetes mellitus. Chronic kidney disease with CrCl > 60 mL/min was diagnosed in 21.8% of participants. The baseline characteristics of the study participants, along with laboratory data (kidney function, Hb_{A1c}, lipid profile), are summarized in Table 1. Demographic and clinical characteristics at baseline showed no statistically significant differences between the two groups.

Primary study outcomes

Achieving blood pressure goals according to the JNC-8 guidelines

The baseline and follow-up levels of systolic BP and diastolic BP are detailed in Table 2. At baseline, none of the enrolled patients were achieving the target BP goals, with 65.3% not using the recommended antihypertensive. However, by the end of the study, a significantly higher proportion of patients in the intervention group (79.4%) were on

Table 1 Demographics characteristics of study participants ($n = 142$)

Patient characteristic		Intervention group, $n = 68$ (%)	Control group, $n = 74$ (%)	p-value*
Age (years), mean ± SD		68 ± 11.26	74 ± 10.31	0.57
Gender	Male	42 (61.8)	45 (60.8)	0.81
	Female	26 (38.2)	29 (39.2)	
Education	Illiterate	12 (17.6)	8 (10.8)	0.33
	High school	27 (39.7)	34 (45.9)	
	College	14 (20.5)	19 (25.6)	
	University	15 (22.1)	13 (17.5)	
Smoking status	Active smoker	18 (26.5)	24 (32.5)	0.44
	Non-smoker (quit smoking > 6 months)	50 (73.5)	50 (67.5)	
Body mass index (BMI) category (kg/m^2)	Underweight	1 (1.5)	0 (0.0)	0.71
	Normal	9 (65.8)	15 (20.2)	
	Overweight	15 (2.6)	21 (28.4)	
	Obese	35 (51.5)	32 (43.2)	
	Morbid obesity	8 (11.8)	6 (8.1)	
Comorbidities	Resistant Hypertension only	4 (5.8)	7 (9.5)	0.43
	Diabetes mellitus	39 (57.3)	44 (59.4)	
	Ischemic heart disease	52 (76.4)	54 (72.9)	
	Heart failure	13 (19.1)	10 (13.5)	
	Chronic kidney disease	16 (23.5)	15 (20.3)	
Laboratory results baseline, mean ± SD	Serum creatinine (mg/dl)	1.1 ± 0.53	1.02 ± 0.38	0.30
	Glycosylated hemoglobin (HbA1c)	7.33 ± 2.21	7.56 ± 2.42	0.55
	Low-density lipoprotein cholesterol (LDL-C (mg/dl))	114.67 ± 41.93	121.18 ± 43.4	0.38

*Calculated by chi-square test or independent sample t-test as appropriate

Table 2 Baseline and follow-up blood pressure (BP) clinical characteristics comparing both study arms

BP Baselines (mmHg)		**Intervention group, $n = 68$, mean ± SD**	**Control group, $n = 74$, mean ± SD**	***p*-value***
Baseline	Systolic BP	157.07 ± 11.11	157.18 ± 10.05	0.94
	Diastolic BP	94.48 ± 4.24	94.77 ± 5.30	0.72
The absolute decline in SBP and DBP (mmHg) from the baseline after 4-, 8-, and 12-week follow-up				
Baseline – 4 weeks	Systolic BP	24.73 ± 12.58	23.14 ± 13.31	0.464
	Diastolic BP	14.12 ± 6.92	12.00 ± 7.94	0.094
Baseline – 8 weeks	Systolic BP	26.60 ± 13.50	24.58 ± 11.00	0.329
	Diastolic BP	14.87 ± 6.47	12.10 ± 7.64	**0.02**
Baseline – 12 weeks	Systolic BP	27.90 ± 11.70	20.40 ± 9.80	**< 0.001**
	Diastolic BP	13.40 ± 6.10	9.00 ± 6.40	**< 0.001**

*Calculated by independent sample t-test

guideline-concordant medication and achieved BP goals compared to the control group (p-value = 0.003).

The analysis revealed that the presence of chronic kidney disease (CKD) and chronic nonsteroidal anti-inflammatory drugs (NSAIDs) use had significant negative effects on BP control. Patients without CKD or chronic NSAID use had better BP control, while other variables did not significantly affect BP control (Table 3).

Clinical pharmacist interventions

Patients in the intervention group received various pharmaceutical services, including education, counseling, and adherence support, resulting in increased patient trust and satisfaction. Patient education on pharmacological and non-pharmacological management, along with adherence counseling, was the most frequent intervention during

Table 3 The correlation between different variables and achieving blood pressure (BP) control in treatment-resistant hypertension

Variables		Achieve target BP, $n = 142$ (%)		p-value*
		Yes	No	
Gender	Male	62 (71.3)	25 (28.7)	0.201
	Female	33 (60.0)	22 (40.0)	
Diabetes mellitus	Yes	53 (64.0)	30 (36.0)	0.373
	No	42 (71.0)	17 (29.0)	
Ischemic heart disease	Yes	72 (68.0)	34 (32.0)	0.685
	No	23 (64.0)	13 (36.0)	
Heart failure	Yes	17 (81.0)	4 (19.0)	0.208
	No	78 (64.5)	43 (35.5)	
Chronic kidney disease	Yes	15 (48.4)	16 (51.6)	**0.018**
	No	80 (72.0)	31 (28.0)	
NSAIDs use	Yes	9 (42.9)	12 (57.1)	**0.020**
	No	86 (71.0)	35 (29.0)	
Smoking	Yes	28 (63.6)	16 (36.4)	0.700
	No	67 (68.4)	31 (31.6)	

*Calculated by chi-square test

NSAIDs, nonsteroidal anti-inflammatory drugs

the first visit, accounting for 51.9% of total therapeutic recommendations.

A total of 158 interventions were identified, averaging 2.49 ± 1.01 TRPs per patient. These included patient education and adherence counseling (51.9% of TRPs), efficacy issues (32.3%), safety issues (8.3%), and miscellaneous interventions like starting single pill combinations and changing drug administration timing.

Declined recommendations included efficacy-related (ACE inhibitors or ARBs preferred over methyldopa or hydralazine), additional combination therapy needs, and patient safety (monitoring for adverse drug reactions).

Secondary study outcomes

Patients' adherence, satisfaction, and QoL

At baseline, adherence scores were comparable between groups. Patients in the intervention group demonstrated significantly improved adherence to medications and all lifestyle measures compared to the control group after the follow-up period ($p < 0.001$). Physical activity levels did not change significantly, but two intervention participants reported increased activity (Table 4). Four patients purchased BP monitoring devices, indicating increased health concerns due to close monitoring. Moreover, patient satisfaction levels with clinical pharmacist interventions significantly increased over the study period, while it was higher in the control group at baseline. Additionally, patients in the intervention group showed improved QoL compared to baseline and the control group ($p < 0.001$).

Physicians' satisfaction and attitudes regarding clinical pharmacist services

Physicians expressed overall positive attitudes toward clinical pharmacist services and were highly satisfied with the activities offered by the clinical pharmacist. They unanimously agreed on the importance of clinical pharmacists in patient education, treatment monitoring, side effect management, drug information provision, and suggesting treatment plan changes. All physicians were satisfied with all activities offered by the clinical pharmacist. Furthermore, 50% of physicians expressed disagreement with the statement, "Clinical

Table 4 Comparison between the intervention and control groups in mean scores of patient's adherence, satisfaction, and quality of life (QoL) at baseline and follow-up ($n = 142$)

Study outcome	**Baseline, mean ± SD**			**Follow-up, mean ± SD**		
	Intervention ($n = 68$)	**Control ($n = 74$)**	**p-value***	**Intervention ($n = 68$)**	**Control ($n = 74$)**	**p-value***
Adherence	6.00 ± 2.28	5.97 ± 1.98	0.94	2.82 ± 1.5	5.95 ± 2.01	**< 0.001**
Medications[1]	1.44 ± 0.5	1.58 ± 0.49	0.097	1.83 ± 0.37	1.58 ± 0.49	**< 0.001**
Exercise[1]	1.18 ± 0.38	1.09 ± 0.29	0.154	1.20 ± 0.41	1.09 ± 0.29	0.063
Diet[2]	3.65 ± 1.32	3.79 ± 1.13	0.468	1.41 ± 0.93	3.77 ± 1.18	**< 0.001**
Satisfaction[3]	3.69 ± 1.27	4.50 ± 1.28	**0.005**	11.89 ± 0.30	4.54 ± 1.35	**0.0001**
QoL (score from 1 to 5; higher scores mean higher QoL)						
General health	3.18 ± 0.79	3.17 ± 0.83	0.99	4.10 ± 0.76	3.16 ± 0.083	**< 0.001**
Health rating	3.06 ± 0.94	3.04 ± 0.99	0.91	4.30 ± 0.65	3.00 ± 0.55	**< 0.001**

*Calculated by independent sample t-test

[1]For both medication compliance and exercise, a score of 1 means non-compliance, score of 2 means compliance

[2]Patients who have a score of < 3 out of 6 were considered compliant to diet, score of > 3 means non-compliance

[3]For satisfaction score, higher scores mean higher satisfaction

pharmacists should leave patient care to doctors." The other ten (45.5%) were neutral, but the last one (4.5%) agreed. Lastly, the item that states that "It's applicable to implement the clinical pharmacist services in the RMS hospitals" got the agreement of 82% of physicians.

Discussion

The prevalence of TRH varies globally, with estimates ranging from 9 to 18% [21]. In the USA, the prevalence of TRH was 17.7% in 2008 and increased to 19.7% in 2018 [4]. Our study is the first to assess the prevalence of TRH in Jordan, finding it to be 12.6%, a figure comparable to the US prevalence. This suggests that TRH is a significant public health issue in Jordan, requiring effective management strategies.

Limited studies worldwide have explored the role of clinical pharmacists in TRH management [13, 14, 22, 23]. This randomized controlled trial aims to assess the impact of a collaborative model involving clinical pharmacists and physicians on TRH management in Jordan, with a focus on patient education. This trial demonstrated that the incorporation of clinical pharmacists into the healthcare team significantly improved multiple aspects of TRH management, including BP control, adherence to medications and lifestyle modifications, patient satisfaction, and QoL.

Our intervention group showed a statistically significant higher percentage (79.4%) of patients achieving their BP targets compared to the control group (55.4%). This aligns with findings from Wooley et al. (2016), who reported that 40% of study participants achieved home diastolic BP goals, and Smith et al. (2016), who found 34.2% BP control with physician-pharmacist collaboration versus 25.9% with usual care [13, 23]. Borenstein et al. (2003) also noted that 60% of uncontrolled hypertensive patients in the clinical pharmacist co-management group achieved BP control compared to 43% in the physician-only group ($p = 0.02$) [24]. These results underscore the critical role of clinical pharmacists in achieving better BP control and supporting the effectiveness of team-based care in managing TRH.

At baseline, only one-third (34.7%) of participants were on the recommended antihypertensive combination per JNC-8 guidelines. Post-intervention, this increased to 76.1%, with a higher percentage in the intervention group (68%) compared to the control group (41%). This improvement is comparable to Albsoul-Younes et al. (2011), who reported that six months into their study, 50% of patients in the intervention group were on the recommended drugs versus 43.4% in the control group [15]. The reduction in BP at the 12-week follow-up was significant. The intervention group saw an absolute decline compared with the control ($p > 0.001$). Similar reductions were reported in the literature [25–27]. These reductions in BP are clinically significant, as sustained reductions of 7 mm Hg in systolic BP can reduce cardiovascular mortality by 10–20% and all-cause mortality by 8–10% [25].

Furthermore, our study emphasized the role of clinical pharmacists in promoting patient adherence to antihypertensive medications and lifestyle modifications (diet and physical activity). Improved adherence to antihypertensive medications has been previously linked to better BP control, consistent with previous findings [14].

In our study, physicians agreed with and implemented 86.8% of the clinical pharmacist's therapeutic recommendations, with only 9.2% being disagreed upon. This high acceptance rate is consistent with Albsoul-Younes et al. (2011), who reported a 67.4% acceptance rate of pharmacist recommendations for uncontrolled hypertension. These findings indicate a high level of trust and collaboration between physicians and clinical pharmacists. This sizable acceptance and implementation rate of physician-approached recommendations by the clinical pharmacist reflected the physicians' acceptance of the clinical pharmacist as a valuable team member. Previous studies also support this finding, noting that physicians felt more secure about prescribing with a clinical pharmacist available for consultation [28, 29].

The response rate to the physicians' satisfaction and attitudes questionnaire was 100%, higher than similar studies in Saudi Arabia (75.5%) and Jordan (66.67%) [30]. Almost all of the surveyed physicians were satisfied with the services offered by the clinical pharmacist, a promising indication for the future integration of these services. This aligns with Tahaineh et al. (2019), who reported high levels of physician comfort with clinical pharmacy services [31]. Conversely, Bilal et al. (2016) found healthcare providers in Ethiopia were dissatisfied due to limited clinical pharmacist availability [18].

Patient satisfaction is also an important aspect of healthcare quality. In this study, the intervention group exhibited higher levels of satisfaction with clinical pharmacist services. The improved patient satisfaction can be attributed to the clinical pharmacist's ability to provide comprehensive drug information, address patient concerns, and tailor medication regimens to individual patient needs [32]. Similar findings were reported emphasizing the role of effective communication and collaboration between healthcare professionals, including clinical pharmacists, in enhancing patient satisfaction [33, 34].

Moreover, the study demonstrated that clinical pharmacists can contribute to improving patients' QoL. Patients in the intervention group reported significantly higher scores in both general health and health rating domains, reflecting the broader impact of clinical pharmacist interventions on overall well-being. The QoL improvements observed in this study echo the prior findings of another study, reinforcing the potential of clinical pharmacist services to enhance the quality of care and patient experiences [35].

Strength and limitations

The main strength of this study is its focus on TRH, a group of patients not commonly the focus of research. The study has several limitations. It was conducted in a single-center setting, which may limit the generalizability of the findings to other healthcare settings. Additionally, the lack of blinding may lead to potential bias due to the social relationship between physicians and clinical pharmacists. The follow-up period was also relatively short, which may have influenced the long-term sustainability of the observed improvements. A study with a larger patient number and longer follow-up period is recommended in the future.

Practical implications and recommendations

The study's results underscore the need for the integration of clinical pharmacists into multidisciplinary healthcare teams. Clinical pharmacists can play a pivotal role in optimizing the management of TRH, improving BP control, enhancing adherence, and ultimately enhancing patient outcomes and healthcare quality. Healthcare institutions should consider integrating clinical pharmacists into hypertension management teams, particularly for TRH patients. Clinical pharmacists should prioritize patient education, providing clear information on medications, lifestyle modifications, and the importance of adherence. Multiple educational sessions over time may further enhance patient understanding and adherence.

Conclusions

In conclusion, the results of this study demonstrate the significant contributions of clinical pharmacists in the management of TRH. The collaborative approach between clinical pharmacists and physicians significantly improves BP control, adherence, patient satisfaction, and QoL. These findings emphasize the importance of integrating clinical pharmacists into the healthcare team to enhance patient outcomes and healthcare quality.

Author contribution All authors have contributed equally, read, and agreed to the published version of the manuscript.

Data availability The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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